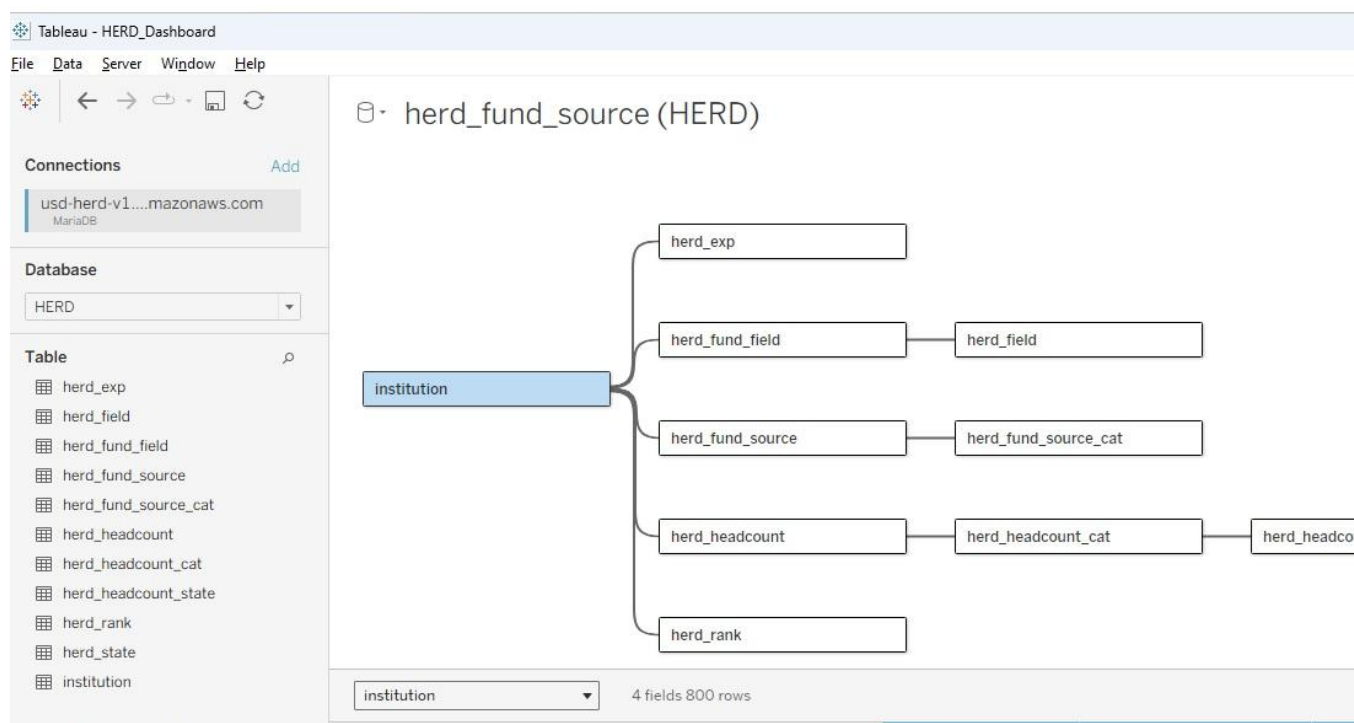


Create HERD Dashboard with Tableau/MariaDB Database

Connect to Maria Db Database

1. Open Tableau
2. Under “Connect” and “To a Server” on the left hand side, click MariaDB
3. Connect to Maria Db Server using “Username” and “Password” if required.
4. Join the related datasets as shown in image below (Unique identifier for join is “Inst ID” for first five and the only “ID” field for the remaining tables :



Build Dashboard in Tableau

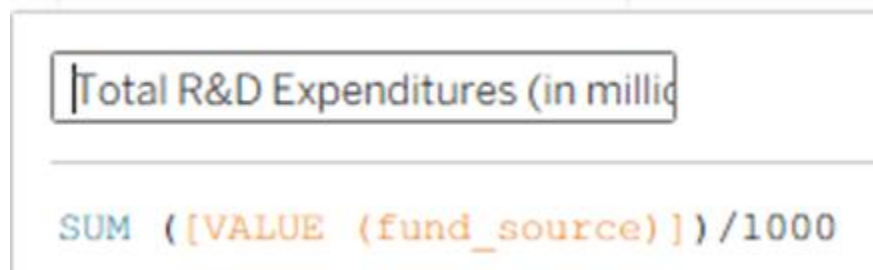
The graphs and dashboards an institution chooses to create will vary depending on institutional need. The following section provides some information and tips about using the HERD data to create graphs in Tableau.

1. List of fields used in primary graphs: <https://www.sandiego.edu/ire/usd-research/herd-dashboards.php>

Type of Graph	Table Name	Field Name
All Graphs	institution	Inst Name

R&D Expenditures by Fund Source	herd_fund_source	VALUE (fund source)
R&D Expenditures by Fund Source	herd_fund_source	Fiscal Year
R&D Expenditures by Fund Source	herd_fund_source_cat	Fund Source
R&D Expenditures by Academic Discipline (Field)	herd_field	Field Name
R&D Expenditures by Academic Discipline (Field)	herd_fund_field	VALUE (herd fund field)
R&D Expenditures by Academic Discipline (Field)	herd_fund_field	YEAR
R&D Personnel	herd_headcount	Headcount
R&D Personnel	herd_headcount	FTE
R&D Personnel	herd_headcount	YEAR (herd headcount)

2. Change number format of the two total R&D Expenditures measures to millions:
 - a. In the “herd_fund_source” table, the “VALUE (fund source)” field is the total amount of R&D Expenditures for all fund sources and has a format of (in millions). Which means a value of 1,500 is actually 1.5 million.
Note. In the “herd_fund_field” the “VALUE (herd fund field)” is the total amount of R&D Expenditures for all fields (academic disciplines) and also has a format of (in millions).
 - b. To convert the “VALUE (fund source)” field to millions, use the following formula to create a Measure field in Tableau named “Total R&D Expenditures (in millions): Can repeat this process with Value (herd fund field)



3. Institutions may want to display a graph with the Total R&D Expenditures split into two categories “External Funds” and “Institutional Funds”
 - a. Create a “Fund Source (group)” dimension field in Tableau:

Note. In the “herd_fund_source_cat” table, the “Fund Source” variable contains the following values: Business, Federal government, Institution funds, Nonprofit organizations, State and local government, and all other sources.

- b. To group the external sources separate from “Institution funds”, click on the arrow to the right of “Fund Source” and under “Create”, select “Group”
- c. Group the field values as shown below:

Edit Group [Fund Source (group)]

Field Name: Fund Source (group)

Groups:

External funds

All other sources

Business

Federal government

Nonprofit organizations

State and local government

Institution funds

- 4. Create a “Peer Comparison” Parameter that will allow users to toggle between graphs that show their institution compared to peer/aspirational institutions.
 - a. Create a parameter named “Select Peer Group” Users will be able to choose “Peer Group” or “Aspirant Group” and the graphs on the dashboard will display totals for whichever group is selected.

Edit Parameter [Select Peer Group:]

Name

Select Peer Group

Properties

Data type

String

Display format

USD Aspirant Group

Current value

USD Aspirant Group

Value when workbook opens

Current value

Allowable values

☐ All

☒ List

☐ Range

Value	Display As
USD Peer Group	USD Peer Group
USD Aspirant Group	USD Aspirant Group
Click to add	

☒ Fixed

☐ When workbook opens

Add values from

- b. Add “Inst Name” as a filter on a sheet(s) and filter the sheet to peer institutions.

- c. Create a calculated field called “Display Sheet” and type the parameter name “Select Peer Group:” in the field. Add “Display Sheet” as a filter. This will display the graph when “Peer Group” is selected on the “Select Peer Group” filter.

DisplaySheet

[Select Peer Group:]

- d. Repeat steps b-c on the sheets you want to filter to aspirant institutions.
 - e. Add the “Select Peer Group” parameter to the dashboard.
 - f. Use sheet swapping on the dashboard by adding a vertical container and adding both graphs (one filtered to peer institutions and one filtered to aspirational institutions). Change the “Fit” of the graphs to “Entire View”
Note. Only one graph will show at a time; the one that is selected in the “Select Peer Group” parameter.
5. Create a Parameter that will allow users to toggle between Headcounts and FTEs on a Personnel graph
- a. Create a measure field in Tableau named “Metric Choice” with the following formula:
Note. “Headcount” and “FTE” are located in the “herd_headcount” table.

Metric Choice

IF [Select Metric:] = "Headcount" THEN sum([Headcount])
ELSEIF [Select Metric:] = "FTE" then sum([FTE])
END

- b. Add the “Metric Choice” field to “Rows”
- c. Create a Parameter with the following values named “Select Metric:” and add to dashboard

Edit Parameter [Select Metric:]

Name
Select Metric

Properties

Data type: String Display format: FTE

Current value: FTE Value when workbook opens: Current value

Allowable values

☐ All ☒ List ☐ Range

Value	Display As
Headcount	Headcount
FTE	FTE

☒ Fixed ☐ When workbook opens

Add values from ▼

6. Create a "Parameter" to allow a user to display five, seven, or 10 years of data in a graph.
- a. Create three "Sets" using the "Fiscal Year" measure in the "herd_fund_source" table. Create a set for five years, seven years, and 10 years. Here is an example of the "5 years" set which will display the last five years of data in a graph:

Edit Set [5Yrs]

Name: 5Yrs

General Condition Top

☐ None

☒ By field:

Top 5 by

Fiscal Year Minimum

- b. Create a "Calculated Field" named "Time Period" and add as a filter on a graph with "True" selected

Time Period

```

If [Select Time Period:]="1" then [3Yrs]
Elseif [Select Time Period:]="2" then [5Yrs]
Elseif [Select Time Period:]="3" then [7Yrs]
End

```

- c. Create a "Parameter" named "Select Time Period:" and add to dashboard

Edit Parameter [Select Time Period:]

Name
Select Time Period:

Properties

Data type
String

Display format
3

Current value
7 Years

Value when workbook opens
Current value

Allowable values

☐ All
☒ List
☐ Range

Value	Display As
1	3 Years
2	5 Years
3	7 Years

☒ Fixed
☐ When workbook opens

Add values from

Embed HERD Tableau Dashboard on a website:

Export iframe

From Tableau Public

To generate an iframe that you can embed into your website, do the following in Tableau Public:

1. The Share button appears at the top right of the page
2. Copy the iframe code, which will look something like:

```
<iframe style="box-shadow: 0 0 0 0.5rem #fff, 0 0 0.25rem 0.5rem rgb(0 0 0 / 5%), 0 0.25rem 0.5rem 0.5rem rgb(0 0 0 / 10%);"
src="https://public.tableau.com/views/HERDDashboard/HERD_Data_Dashboard?:language=en-US&;display_count=n&;sid=&;redirect=auth&;&;showShareOptions=false&;origin=viz_share_link:embed=y&;display_count=yes&;showVizHome=no&;showShareOptions=false" width="100%" height="1100px" frameborder="0"></iframe><hr />
```

From the USD Website

You can also use an already-existing chart on the USD website and embed that into your own site. To do this:

1. Load the page with the Tableau chart that you'd like to embed, for example:
<https://www.sandiego.edu/ire/usd-research/herd-dashboards.php>
2. Scroll and click on the "View in Tableau Public" link on the bottom left
3. The dashboard will open in Tableau Public
4. The Share button appears at the top right of the page
5. You can either copy the iframe provided, or you can also append the following syntax to the link URL; this will strip the Tableau Public header from the top of the page, so only the dashboard shows up in the iframe with a reference to Tableau:

`https://public.tableau.com/views/HERDDashboard/HERD_Data_Dashboard?:embed=y&;display_count=yes&;showVizHome=no&;showShareOptions=false`

Embed via your CMS

This process will vary depending on the Content Management System (CMS) used by your organization.

Cascade Server

If you try to paste an iframe directly into the visual editor, Cascade might delete it when you hit "Submit" because it's not "valid XHTML." Instead, place the code directly into the source code of any text field for display. Ensure the iframe tag is "self-closing" or has a proper closing tag: `</iframe>`. If it's just `<iframe>`, Cascade's XML parser will throw an error.

Drupal

Drupal's "security-first" approach means it will often "strip" (delete) iframe tags when you save a page. When editing a page, change the Text Format dropdown (usually below the body field) to "Full HTML". If it still doesn't work, an admin needs to go to Configuration > Content Authoring > Text Formats and ensure the `<iframe>` tag is allowed in the filter settings.

Experience Cloud

Since Salesforce owns Tableau, they have a native "Tableau Viz" lightning component. Don't use a manual iframe if you can avoid it. Drag the Tableau Viz component onto the page in the Experience Builder. It handles authentication (Single Sign-On) much more easily than a raw iframe.

Hubspot

Use a "Rich Text" module, click Insert > Embed, or edit the Source Code (`<>`) to paste the iframe directly.

Wix

Go to Add (+) > Embed Code > Embed HTML. In the "What's in the HTML?" box, paste your iframe code.

WordPress

Use the "Custom HTML" block. If you are using the classic editor, switch to the "Text" tab (instead of Visual) and paste the code.